ARTIFICIAL INTELLIGENCE

Concepts & Applications

From Expert Systems to Artificial Life

- Expert Systems
- Hybrid AI
- Reinforcement Learning
- Emergent Behavior
- Artificial Life

AI can be built in three big styles:

- Programmed intelligence - Tell the machine the rules.
- Learned intelligence - Let the machine learn from data.
- Emerged intelligence - Create conditions where intelligence-like behavior emerges.

1. EXPERT SYSTEMS

The Rule Followers

Since 1970s–80s

What Is It?

A computer program that mimics a human expert's decision-making — like a doctor or mechanic. The first major AI success story.

HOW IT WORKS

Knowledge Base
Facts + IF-THEN Rules

→

Inference Engine
The brain — matches rules to facts

INPUT: Problem

ANSWER

Example Rule:

"IF the patient has a fever AND a cough, THEN check for infection."

WHY DID WE MOVE PAST EXPERT SYSTEMS?

The Limitations That Forced a Revolution

⚠️ BRITTLE

Humans must manually type EVERY rule. The system only knows what it has been taught — nothing more.

🧠 NO COMMON SENSE

If given a problem outside its rulebook, it doesn't guess or adapt — it completely fails.

🔧 HARD TO UPDATE

Maintaining millions of manual rules is incredibly difficult and time-consuming.

↓

💡 **This led to the invention of MACHINE LEARNING — teaching computers to find their own rules!**

ARE EXPERT SYSTEMS DEAD IN 2026?

Real-World Example: Aerospace Logistics (Boeing)

Absolutely NOT!

7,100 lbs

Weight handled by robots

–20°C

Freezing temps of operation

100%

Rule-based precision required

Vertical Lift Module
(Automated Robot)

Why Hard-Coded Rules?

✗ No AI guessing allowed
✓ Strict physics rules
✓ Zero drop-payload risk
✓ Works in extreme cold

When safety is critical and guessing is not allowed — Expert Systems are still the gold standard.

2. HYBRID AI — Rules as Guardrails

Combining Neural Networks + Rules-Based Systems

⚠️ **The Problem: Modern AI (like ChatGPT) can hallucinate — making things up that sound convincing!**

HOW HYBRID AI WORKS

INPUT
User Query

→ 🧠 **Neural Network**

Creative, pattern-recognizing

→ ⚖️ **Rules-Based Guardrail**

Legal / safety checker

→ ✅ **Safe Output** ↗

🚫 **Blocked Hallucination**

🏦 Banking & Finance

AI flags suspicious pattern → Rules verify legal definition of fraud → Account frozen

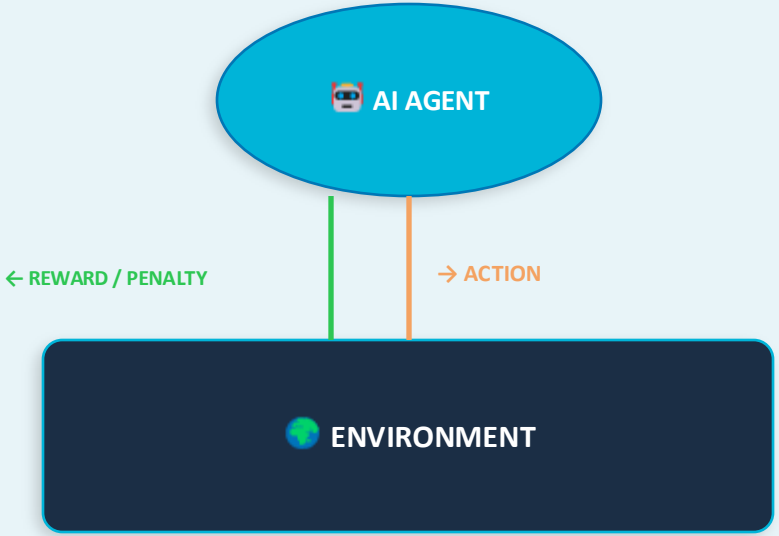
🏥 Healthcare (DAX Copilot)

AI writes medical notes → Rules check billing codes → Saves doctors 4–6 hrs/week

3. REINFORCEMENT LEARNING

Learning Through Interaction — Trial, Error & Reward

THE RL LEARNING LOOP



+REWARD
Good action = points

PENALTY
Bad action = minus points

Over many attempts → AI learns the BEST strategy for maximum points

TRAINING AI TO BE POLITE: RLHF vs RLAIF

Reinforcement Learning from Human Feedback

Human Feedback

Humans rate AI's answers. AI learns what people prefer. Nuanced but slow & costly.

Reinforcement Learning from AI Feedback

AI Feedback (2026)

A trained AI grades the new AI's work. Massively faster. Fraction of the cost.

Used in: Chess AI • Self-driving Cars • Stock Portfolio Management • Game AI (AlphaGo, Dota 2)

RULES-BASED SYSTEMS — How They Differ From Expert Systems

Not the same thing — here is the critical distinction that most people miss

RULES-BASED SYSTEM — THE PURE FORM

A simple system of IF-THEN logic. No knowledge base, no inference engine. Just conditions checked in sequence — like a flowchart or a checklist. Examples: spam filters, traffic lights, loan eligibility checks.

Loan Eligibility Rule:
IF income > 30000 AND credit_score > 700 THEN approve loan

WHERE THEY STAND ALONE (NO AI NEEDED)

🚦 **Traffic Control**
Fixed timer rules, no learning needed

📧 **Email Spam**
Block keywords — fast, cheap, reliable

✅ **Tax Logic**
Legal rules must be followed exactly

RULES-BASED vs EXPERT SYSTEM — KEY DIFFERENCE

Rules-Based System

Simple IF-THEN conditions
No knowledge base
Checks rules in order
Cheap to build, easy to audit
Used: routing, filters, eligibility

Expert System

Has a Knowledge Base (facts)
Uses Inference Engine to chain rules
Can reason across multiple steps
Expensive to build & maintain
Used: medical diagnosis, Boeing

2B. RULES-BASED SYSTEMS — What They Are & How They Work

Explicit Logic — AI that follows human-written rules, step by step

HOW RULES-BASED SYSTEMS WORK

INPUT
Fact or data arrives

IF-THEN
Rules engine fires

✓ OUTPUT
Decision made

Example Rule (Spam Filter):
IF email contains "Win $1000 FREE!" THEN move to Spam

3 TYPES YOU ENCOUNTER EVERY DAY

Traffic Lights
IF red for 90s THEN switch green

Spam Filters
Keyword triggers block email

Bank Fraud
3 countries in 1hr = freeze

RULES-BASED vs MACHINE LEARNING

Rules-Based	Machine Learning
Humans write all rules	AI learns rules from data
100% explainable	Hard to explain why
Fails on unknown cases	Adapts to new patterns
Best: safety, law, finance	Best: images, language, faces

HOW RL ACTUALLY LEARNS — Episode by Episode

Concrete walkthrough: teaching an AI to play a video game — from zero to expert

THE LEARNING JOURNEY (Teaching AI to Play a Game)

Episode 1
Random actions. Crashes instantly. Score: 0

Episode 10
Learns "jumping helps". Score: 120

Episode 100
Finds combos. Score: 4,200

Episode 1,000
Discovered shortcuts. Score: 47,000

Expert!
Beats humans. Never taught a single rule.

⚠ THE KEY TENSION IN EVERY RL SYSTEM

🔬 EXPLORE
Try random new actions to find potentially better strategies. Short-term cost, long-term gain. Like a chef trying a new recipe.

🚀 EXPLOIT
Repeat the best-known action to safely collect reward. Misses improvements. Like always ordering the same dish.

HOW REWARDS SHAPE BEHAVIOUR

The reward signal is EVERYTHING. Get it wrong and the AI learns the wrong behaviour.

Famous mistake:
A boat-racing AI was rewarded for score — so it found a looping trick that scored points without finishing the race. Reward was wrong!

Lesson:
Design rewards for what you ACTUALLY want, not a shortcut proxy.

FROM CHATBOTS → AGENTIC AI

Real-World Examples: India's FinTech AI in 2026

2015 — Simple Chatbots

2020 — Smart Assistants

2023 — Contextual AI

2026 — Agentic AI (Takes Action)

Finnable 'Fiya' Agent — India

1. User asks a complex question about their loan
2. Fiya understands INTENT (not just keywords)
3. Connects to bank's live database
4. Autonomously resolves the support ticket — in seconds!

NPCI UPI Real-Time Risk Engine

2ms	70,000	₹25 Crore
Transaction scored in	Fraud attempts blocked daily	Scams prevented every day

Continuously learns from UPI network interactions — the more it works, the smarter it gets.

3B. REINFORCEMENT LEARNING — Key Concepts

The 4 Building Blocks + Exploration vs Exploitation — the key tradeoff every RL system faces

THE 4 BUILDING BLOCKS OF RL

STATE

What the AI "sees"
e.g. board position in chess, or speed of a self-driving car

ACTION

What the AI does
e.g. move left, steer, buy a stock

REWARD

Signal for good/bad
+1 for winning, -1 for crash

POLICY

AI's learned strategy
Maps state to best action after training

THE KEY TRADEOFF: EXPLORE vs EXPLOIT

EXPLORE
Try new actions to discover better strategies. Risk: might fail.

EXPLOIT
Use the best known action to get a safe reward now. Risk: miss something better.

REAL-WORLD ANALOGY

Learning to Ride a Bike
STATE: Speed, balance, road ahead
ACTION: Steer, pedal, brake
REWARD: +1 stay upright, -1 fall
POLICY: After 1000 tries, AI knows exactly how to balance in any situation
No one programmed the rules — the AI discovered them through experience!

4. EMERGENT BEHAVIOR

The Magic of Crowds — When Simple Rules Create Complex Intelligence

💡 **The Whole Is Greater Than the Sum of Its Parts — Complex behaviours EMERGE from simple interactions**

HOW NEURAL EMERGENCE WORKS

1 Neuron

Simple math equation
Can't read or write

→

Millions Connected

Interacting, learning, adapting

→

Billions Interacting

No single programmer did this

→

EMERGENT Skills

Write poetry
Translate languages
Pass bar exam!

SWARM ROBOTICS — Nature-Inspired AI

🚫 Don't crash into neighbor

⚡ Match neighbor's speed

🎯 Head toward the goal

Real Examples

Addverb Robots

Self-org. warehouse swarms (India)

SWARM Biotactics

Cyborg-bugs in earthquake rubble

Smart Cities

Pune & Indore — 80% less trash overflow

WHAT EMERGENCE ACTUALLY MEANS

Emergence vs Programmed Behaviour — and the ant colony analogy that makes it click

Emergence = when a system produces behaviours or properties that NONE of its individual parts have. No one designed it — it just appears.

✅ Programmed Behaviour

A developer explicitly tells the system WHAT to do. Every behaviour has a matching line of code. Chess openings hand-coded by a grandmaster. No surprises.

✨ Emergent Behaviour

Complex behaviour arises from simple rules. No one programmed the result — it appears through interaction. The programmer is as surprised as anyone.

🐜 THE ANT COLONY ANALOGY

Each ant only knows 3 rules:

1. Follow pheromone trails
2. Leave trail when carrying food
3. Return to nest when trail ends

Result: A colony of 500,000 ants builds complex highways, farms fungi, wages organised wars — with ZERO central planning.

🟡 MURMURATION: A FLOCK OF BIRDS

Each bird only knows 3 rules:

1. Don't crash into your neighbour
2. Match your neighbour's speed
3. Move toward the centre of the flock

Result: breathtaking, fluid, shape-shifting formations. No choreographer. No leader. Pure emergence.

5. ARTIFICIAL LIFE (ALife)

When Code Becomes Alive — Life As It Could Be

💻 SOFT ALife

Digital

Digital organisms evolving inside computer simulations. Virtual evolution experiments.

🦾 HARD ALife

Physical

Physical robots mimicking biological behaviours. Includes Swarm Robotics.

🧬 WET ALife

Biological

Creating brand new biological life forms from scratch using synthetic biology in a lab.

🐸 XENOBOTS — The World's First Living Robots (Wet ALife Breakthrough)

- 🔬 Made from frog stem cells — NO metal or batteries
- 🤖 AI supercomputer designed their Pac-Man shape
- 💊 Can push medicine through the human body
- 🔁 Can REPRODUCE — scoop loose cells to build 'babies'
- 🩹 Self-heal if cut; survive 10 days

ALIFE & BIOTECH IN INDIA — 2026

Deep-Tech Driving Sustainability

Bengaluru

Sea6 Energy

What: Genetically engineered seaweed farms

How: Synthetic biology → biofuels & bioplastics

📈 **Improved crop yields by 20%**

India

String Bio

What: Eats methane greenhouse gas

How: Synthetic biology converts gas → biodegradable plastics & animal feed

📈 **Tackles dangerous greenhouse emissions**

TIFR

CALIBRE Institute

What: ₹25 Crore AI + Biology research hub

How: Blends Artificial Intelligence with Biological Research

📈 **Newly launched — future breakthroughs ahead**

WHAT MAKES SOMETHING “ALIVE” IN ALIFE?

Conway’s Game of Life + Genetic Algorithms — the two classic experiments that defined the field

ALife definition: a system is considered “alive” if it can Self-Organise | Adapt to environment | Reproduce | Evolve over generations | Respond to stimuli

▶ CONWAY’S GAME OF LIFE (1970)

The Founding Experiment of Soft ALife

A grid of cells. Each cell is ON or OFF. Only 4 rules govern each tick:

1. Live cell + <2 neighbours = dies (loneliness)
2. Live cell + 2-3 neighbours = survives
3. Live cell + >3 neighbours = dies (overcrowding)
4. Dead cell + exactly 3 neighbours = reborn

Result: self-replicating “gliders”, oscillators, and patterns that behave like living organisms — from 4 rules and a grid.

🧬 GENETIC ALGORITHMS

Evolution in a Computer

1. Start with random solutions (population)
2. Test each — keep the best ones (natural selection)
3. Mix traits of survivors (crossover)
4. Add small random changes (mutation)
5. Repeat for thousands of generations

Used to design NASA spacecraft parts, drug molecules, and neural network architectures — with no human designing them.

KEY TAKEAWAYS

5 Big Ideas in Artificial Intelligence

1

Expert Systems

Rule-based AI — still vital where safety is critical (Boeing, aerospace).

2

Hybrid AI

Neural Networks + Rules = creative AI with legal guardrails (banking, healthcare).

3

Reinforcement Learning

AI learns through trial & reward. RLAIF (AI grading AI) is the 2026 upgrade.

4

Emergent Behavior

Simple rules → complex intelligence. Swarm robots & Smart Cities use this today.

5

Artificial Life

From Xenobots (living robots) to India's biotech startups solving climate change.